

The coupling effect of carbon emission trading and tradable green certificates under electricity marketization in China

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ABSTRACT

To accelerate the low-carbon transition in the energy sector, China is gradually promoting the construction of the national electricity market and carbon market. Carbon emission trading (CET) and tradable green certificates (TGCs) are advocated as promising tools to control carbon emissions. However, the efficacy of these concurrent low carbon regulations remains an open question, especially considering that their implementation schemes are quite different. Therefore, herein, the coupling effect of carbon emission trading and tradable green certificates is examined under electricity marketization in China. An interactive system dynamics (SD) model is established to capture the internal linkages among CET, TGCs and electricity market. Different markets are connected through the investment behaviour of market participants and the price formation mechanism, and the dynamic equilibrium process of multiple markets is analysed based on the current electricity marketization in China. The simulation results based on real-world data in China show that the coordination of TGCs and CET alleviates the pressure of the fiscal deficit of renewable energy and promotes the low-carbon transition in the energy sector, and relevant policy recommendations are put forward. Hopefully, our work can provide a useful reference and insight for low-carbon policy design and energy market construction.

1. Introduction

To combat global climate change, major economies in the world have made commitments to reduce carbon emissions in recent years [1]. By 2022, more than 100 countries announced that they would achieve carbon neutrality by the middle of the 21st century, and the global carbon emission reduction has ushered in a turning point. However, with the weakening of the impact of the COVID-19 pandemic, countries around the world gradually lifted the blockade, and the global economic recovery also made the global energy demand rebound significantly [2–4]. The soaring natural gas prices and increasingly frequent extreme weather in 2021 emphasized the need for coal-fired power generation, which also led to a rebound in carbon emissions. In 2021, the world's carbon emissions were 33.884 billion tons, up 5.6% year on year. As the country with the highest carbon emissions in the world, China's total carbon emissions in 2021 will exceed 10 billion tons, twice as much as the second-place country (the United States). The proportion of carbon emissions in the power industry will reach more than 40% [5–7].

To ensure the implementation of the “dual carbon” goal, the Chinese government has issued a number of guidelines and regulations for the reform of the mechanism and system. The Renewable Energy Electricity Consumption Guarantee Mechanism, which came into force at the beginning of 2020, clearly implements the renewable energy quota system, requires grid operators, distribution companies, power retailers, and large power users to reach the proportion of non-water renewable electricity consumption each year, and clearly requires the proportion of renewable energy electricity from the power consumption side [8–10]. For this purpose, the National Development and Reform Commission and the Energy Administration have formulated the responsibility weight to be consumed by each province year by year [11]. The weight of consumption responsibility on the power side forces the transformation of the power supply structure, boosting the installation of variable renewable energy. Additionally, relevant departments have formulated annual carbon dioxide emission quotas for fossil fuel power plants and implemented the goal of “double carbon” through the construction of a carbon emission trading (CET) market. The national CET market of China was officially launched in July 2021. As the largest

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Abbreviations

SD	System dynamics
CET	Carbon emission trading
TGC	Tradable green certificates
RPS	Renewable portfolio standards

single carbon emission industry, the electric power industry will be included in the carbon emission trading system for the first time, which will help to control carbon emissions of power generation enterprises guided by the carbon market. The establishment of the carbon market has brought new challenges to the reform of the electricity market [12–14]. The integration of the carbon market and electricity market is primarily manifested in three key aspects: 1) Cost characteristics: the external costs associated with carbon emissions resulting from traditional thermal power generation can be passed on through the electricity market. This transmission process forms the core principle of the coordinated development of the carbon market and the electricity market. 2) Price formation: since the carbon price is incorporated into the cost of power generation, fluctuations in the carbon price impact the supply and demand dynamics in the electricity market, along with the resulting clearing prices. 3) Mechanism design: the common goal of the electricity market and carbon market is to accomplish effective low-carbon development. However, inadequately designed market mechanisms can lead to deviations from the optimal re-source mix, resulting in additional social costs.

The reform of China's electricity market has developed rapidly since it was restarted in 2015 [15]. By the end of 2021, spot electricity trading pilot projects were successively carried out in 14 provincial markets. It has accumulated experience in building a electricity market system that combines medium- and long-term transactions with spot transactions and synergizes the interprovincial and intra-provincial markets [16]. The national green power trading pilot was officially launched in September 2021 [17]. As independent trading varieties in the electricity market, green power and green certificates will discover the environmental value of green power in a market-oriented way. In January 2022, the National Development and Reform Commission and the Energy Administration issued the Guiding Opinions on Accelerating the Construction of a National Unified Power Market System, emphasizing the need to optimize the overall design of the electricity market, realize the sharing, mutual aid and optimal allocation of power resources across the country, and accelerate the formation of a unified, open, orderly, safe, efficient and well governed electricity market system [18]. In fact, it has become the consensus of electricity market construction practice around the world to compensate for the environmental value of low-carbon technologies by conducting green power and tradable green certificate (TGC) transactions.

Driven by ambitious energy policy objectives, the carbon market and the electricity market show a trend of interactive and complementary coupling development [19]. Generally, TGCs and CET are considered the two most common tools implemented by policy-makers for the decarbonization of energy sectors in deregulated markets [20]. TGC and CET have interactive effects due to economic and environmental impacts [21]. For market participants with different carbon emission intensities, TGCs and CET markets provide an effective way to restore the environmental value of low-carbon technologies. For instance, when the TGC market mechanism is implemented, renewable energy producers can obtain benefits from electricity market and TGC market, which can facilitate the investment of renewable energy [22]. From the perspective of CET markets, conventional thermal units with excess emissions are supposed to purchase carbon quotas, which will also enhance the investment in low-carbon technologies. Therefore, the simultaneous TGC and CET mechanism can reduce emission and adjust the energy

structure. The further integration of variable renewable energy as well as other low-carbon technologies will change in the supply-demand situation in CET and TGC markets, resulting in fluctuations in TGC prices and CET prices and further impacting electricity price. In practice, the carbon emission quota, carbon tax and renewable energy consumption responsibility weight are divided by market entities and administrative region, while carbon emission right trading, green card trading, and determining the trading subject and trading volume through market dynamics are not limited by the grid region. In other words, the TGC market promotes the transformation of the energy structure on the generation side, while the CET market promotes the development of low-carbon technology by controlling the carbon emissions of traditional energy. It is of vital importance to investigate the above market interactions as well as the corresponding mix policy effects.

Some of the existing studies examine market interaction through theoretical approaches and analytically solve market equilibria. G. Wang et al. [23] evaluate the coordination effect of the TGC and CET markets in China by applying a multi-region equilibrium model. The coordinated markets are proven to be more efficient with well-formed low-carbon transformation goals. H. Dai et al. [24] assess the economic effect of CET implementation in the energy sector based on a computable general equilibrium model. Analysis shows that the renewable energy targets should be aligned with CET. L. Zhang et al. [25] studied the interaction of CET quota allocation scheme design and electricity market operation with an analytical model and suggested that CET quota allocation be based on emission intensity in the electricity industry. However, to ensure the solvability of the model, such methods often overlook or oversimplify some market modelling, and the lack of market details may lead to misrepresentation of the market interaction. In addition, these traditional static methods have difficulty quantitatively comparing and analysing the chain reaction among markets and cannot fully describe the interaction relationship of multiple influencing factors in the system.

Notably, the above market interaction is inherent and can present the following characteristics: 1) In the energy sector, TGCs promote power supply structure transformation, while CET controls the production capacity of traditional energy and promotes technological innovation. When the two systems are actually implemented, it will become a valuable issue to study their coupling effect on China's electricity market. For the complex interaction of parameters in TGC market, CET market and electricity market, it is difficult for traditional methods to quantitatively analyze the influence and change of parameters. 2) Multiple markets are connected by different market participants from both the generation and consumption sides and interact with each other through numerous endogenous and exogenous variables, e.g., TGC price, CET price and the price of the electricity market. The prices in different markets can provide direct signals for investment decisions of market participants and are able to exert significant influence on the energy transition. If there is no reasonable coordination mechanism, the social welfare transfer and interest conflict between different market players will significantly disrupt the efficient operation of the electricity market as well as the carbon market.

With the above motivation, a number of studies have applied numerical frameworks to investigate the mix policy effects and endogenize investment decisions in low-carbon technologies. These studies highlight the interaction of different market participants under various policy regulations and tend to establish a bottom-up model to capture the coupling effect in multiple markets. R. Budi et al. [26] designed a generation expansion planning model based on evolutionary game theory in partially deregulated markets considering carbon emission reduction targets. The strategic behaviours of heterogeneous market players in TGC markets are investigated in Ref. [27], and the herd effect under different low-carbon policies is discussed. M. Anwar [28] et al. analyze the investment decision of multimarket participants in a deregulated electricity market based on an agent-based model

considering the effect of mixed low-carbon policies. While these studies are inspiring, many of them overlook critical components in market operation when exploring the interaction between market behavior and the energy sector. For example, some studies fail to capture the linkage between different market prices, a factor that has been shown to significantly influence the quantification of environmental value and policy evaluation. Moreover, China's carbon market and electricity market construction possess unique characteristics unlike mature foreign electricity markets and carbon emission rights markets. Notably, the latest batch of Spot market pilots, national carbon market construction, and green power trading pilots in China were all launched after 2020, marking the inception of China's carbon market. In contrast, most previous studies were based on mature foreign markets or only assessed the potential benefits of implementing relevant mechanisms in China from a macro-level perspective, offering limited guidance for the country's present situation. Consequently, the existing studies do not provide a clear understanding of the evolution and conduct of the externality cost for carbon emissions in China's electricity-carbon coupling market. Conducting research based on China's current market construction and actual data holds paramount significance.

To address the aforementioned research gap, we conduct an investigation into the interaction among TGCs, CET, and the electricity market in China. We quantitatively analyze the coupling effect of these components by examining market behavior changes and social welfare transfers. The contributions of our study can be summarized as follows:

- 1) We establish an interactive System Dynamics (SD) model to capture the internal linkages among subsystems, specifically the CET market, TGC market, and electricity market, based on the latest market situation and relevant mechanisms in China. Despite relying on certain assumptions and simplifying certain market processes, the proposed interactive SD model effectively considers two crucial aspects in evaluating low-carbon transformation policies: the price linkage mechanism and the transfer of carbon emissions costs across multiple markets. Through adjustments of boundary conditions and configuration of cost parameters, our model can simulate the interaction between TGC, CET, and the electricity market across various regions, thus facilitating effective collaboration among different low-carbon policies within diverse market environments.
- 2) The interconnection of different markets is facilitated by market participants' investment behavior and the price formation mechanism. We conduct an analysis of the dynamic equilibrium process involving TGCs, CET, and the electricity market, while also refining the driving factors behind carbon emissions in the power industry. Utilizing the most recent datasets and parameters from the electricity and carbon marketization construction in China, our simulation results quantitatively demonstrate the impact of TGC and CET on the electricity markets over the next 10 years, as well as their contribution to the low-carbon transition in the energy sector. Furthermore, we reveal the price linkage mechanism and the mode for transferring carbon emissions costs across multiple markets.
- 3) The case studies, based on real-world datasets, show that the coupling mechanism accelerates the energy transition in most scenarios. However, it may result in higher social costs, and this coupling effect is influenced by the power supply structure in different regions, reinforcing the significance of tailored low-carbon policies according to local conditions. Additionally, parameter uncertainty does not affect the overall trend of energy transformation but alters specific policy effects. In summary, diversified low-carbon policies and carbon-electricity coupled markets can enable power companies to achieve better transformation under emission reduction targets.

Hopefully, our work can provide a useful reference and insight for emission reduction policy design and energy market construction.

2. The coupling effect of CET and TGC

The fundamental aspect of various low-carbon policies, such as CET and TGC, involves redistributing the external environmental costs among diverse market participants. These two policies, namely, CET and TGC, impact renewable energy enterprises and traditional thermal power enterprises, respectively. They facilitate the green low-carbon transformation of the energy industry by altering the supply and demand dynamics within the electricity market. Fig. 1 illustrates the schematic representation of the coupling effect between CET, TGC, and the electricity market.

As shown in Fig. 1, the TGC mechanism offers supplementary environmental value benefits for electricity generated from renewable resources. Likewise, the CET mechanism imposes external environmental costs on carbon emissions produced by traditional coal-fired units. The cost-benefit analysis of relevant market entities considers the advantages of green certificates trading and the costs associated with carbon emissions. Consequently, both the TGC and CET mechanisms can influence the market behavior of these entities, thereby impacting the supply and demand dynamics within the electricity market and the resulting clearing prices.

Regarding the CET market, the allocation of quotas in China's carbon market primarily utilizes a benchmark-based free quota system, with the total quota amount being determined through a bottom-up approach. The process involves two primary steps: firstly, the provincial ecological and environmental departments allocate quotas for each unit based on the actual output, quota allocation method, and carbon emission benchmark values of key emitting units in the region between 2019 and 2020, which are then aggregated to form provincial quotas; secondly, the quotas at the provincial level are summed to establish the total national quota. The CET mechanism enables enterprises to meet their carbon emission reduction targets through both independent emission reduction and market trading. Specifically, based on the government-issued free carbon emission quotas, if an enterprise's actual emissions are lower than the quota, the surplus amount can be sold. Conversely, if the emissions exceed the quota, the enterprise must acquire additional quotas in the CET market. In terms of quota allocation, the future trend will involve a gradual transition from a free quota system to a paid distribution method, with the proportion of paid distribution increasing over time. Additionally, with regard to trading models, there will be a progressive expansion from a sole method of quota spot trading to encompass multiple trading models, including national certified voluntary emission reduction (CCER) and other trading variants.

Overall, the prices observed across multiple markets, namely CET, TGC, and the electricity market, serve as direct signals for market participants when making investment decisions. The transfer of social welfare and conflicts of interest among different market players will impact the operational efficiency of both the electricity market and the carbon market. In terms of policy objectives, the goals of the TGC and CET mechanisms align, as they are both intended to address environmental externalities, optimize resource allocation, and facilitate low-carbon transformation within the energy sector while minimizing social costs.

3. SD model formulation

In this section, the impact on the entire electricity market of the parallel implementation of TGCs and CET is analysed by a complex SD model composed of three subsystems: 1) CET market, 2) TGC market and 3) electricity market. SD is a discipline of analysing information feedback, which can provide real-time feedback and adjust the system parameters and structure based on the actual observed information, so as to seek the most perfect system behaviour mode. Compared with other research methods, the SD model applies to the complex time-varying system with high-order, multi-variable, multi-loop nonlinear and feedback structure, and is able to quantitatively analyze the long-time

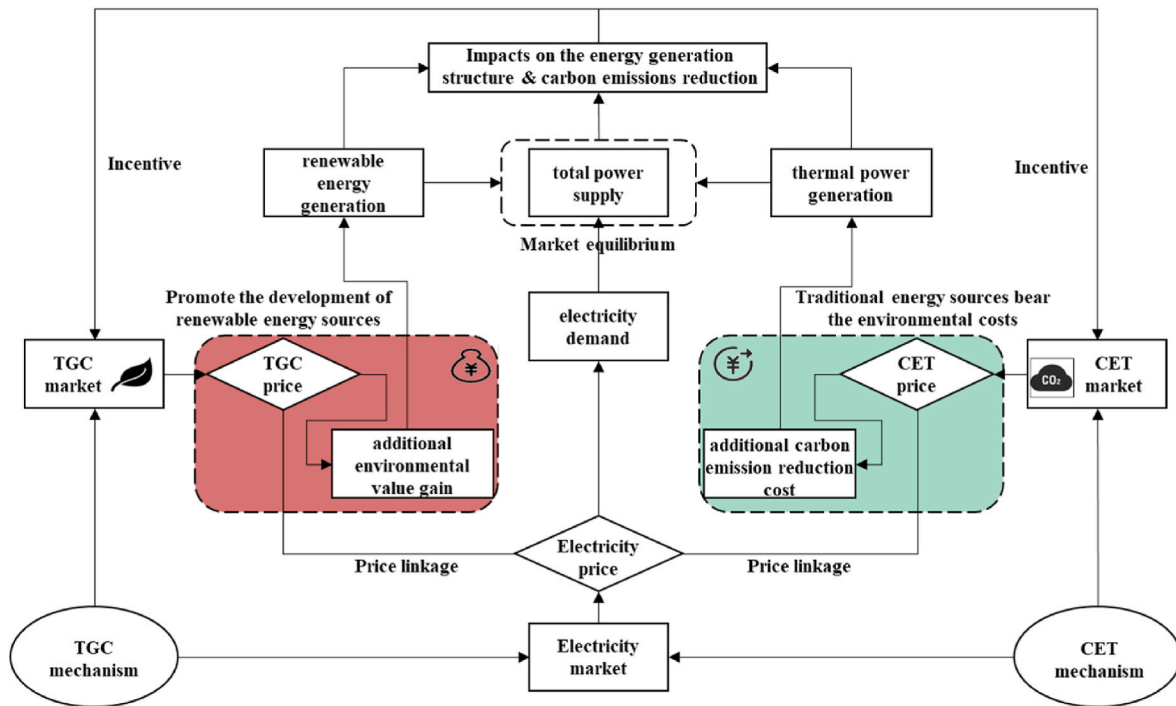


Fig. 1. The schematic of the coupling effect of CET and TGC as well as the electricity market.

impact of each element on the overall system. In this context, SD can show the effect of different quota policies after the implementation of TGCs and CET and the impact on the electricity market. As a result, we use the SD model to investigate the multi-system coupling effect of TGCs and CET on the electricity market, and the interaction between the whole system and each subsystem is elucidated.

3.1. Model framework

1) The main body in the SD model is the generation side. The main participants in the generation side are divided into traditional thermal power generators and renewable energy generators. Traditional thermal power generators use fossil fuels and renewable energy generators use renewable energy. Both the TGC market and the CET market are affected by the behaviour patterns and market transactions of the two generator manufacturers, and the changes in product prices in their respective markets are used as incentive signals to affect the trading behaviour of the electricity market.

Specifically, traditional thermal power generators and renewable energy generators are power suppliers in the system and do power generation and power supply work. In addition, traditional thermal power generators also participate in the CET market in the power generation process to reduce carbon emissions. The seller's power suppliers with lower carbon emissions can sell excess carbon quotas, while the buyer's power suppliers with higher carbon emissions can buy corresponding carbon quotas gaps in CET market.

2) After the 'evaluation method of renewable portfolio standards (RPS)' is approved by the government, the TGC market closely related to the electricity market will be derived. Each renewable energy generator has to reach the quota ratio required by the state to achieve the "30-60" carbon-neutral target. renewable energy generators supply green certificates, which produce a corresponding green certificate for each unit of electricity generated. However, traditional thermal power generators are the demanders of green certificates. thermal power generators cannot produce green certificates themselves, and

they can only meet the quota requirements by purchasing sufficient green certificates in the TGC market.

3) The TGC market and CET market are similar in market operation, and their investment preferences for new units are closely related to commodity price fluctuations. The total installation capacity is obtained by the cumulative effect of the new installation capacity, so the price fluctuation affects the total installation capacity by affecting the new installation capacity, which further affects the development trend of the power generation side over a long-time scale. The prices of both markets are affected by the market supply and demand, but for the carbon emission trading market, the total amount of emissions is set by the government. Therefore, in a sense, the overall trend of the market price of carbon emission rights is controlled by the policy designers

3.2. Model design

3.2.1. Hypothesis

For the convenience of research, we make some assumptions as follows:

- 1) Generators are simply grouped into two categories: renewable energy generators using renewable energy and thermal power generators using traditional fossil energy.
- 2) The market participants on the generation side cannot directly trade green certificates and carbon quotas themselves. Additionally, power suppliers and terminal power users cannot directly trade electricity. Power suppliers need to trade electricity, green certificates and carbon quotas with grid operators.
- 3) CET, TGC and electricity are traded separately in the three markets.
- 4) Each market member declares the real cost and does not speculate.
- 5) The benchmark price of CET in China's carbon market is set at 50 yuan/ton. And the upper and lower limits of CET price are 300 yuan/ton and 10 yuan/ton based on the EU's carbon trading prices.
- 6) The set of new units from the start of construction to actual production needs to experience 12 months of construction time.

- 7) As China's GDP grows, electricity consumption and carbon emissions will also change at a fixed average rate. Among them, GDP growth ratio is based on previous year's data forecast. Similarly, the RPS ratio will increase in accordance with the renewable energy development plan.
- 8) Renewable energy generators will be issued a green certificate by the government for each 1 MWh output. Assume that the green certificate is valid for one month. And green certificate trading is separate from electricity trading.
- 9) There is no difference in the green certificates issued for different types of renewable energy.

3.2.2. CET market

In the CET market, the government limits the maximum value of carbon emissions to achieve the “30-60” carbon-neutral target, resulting in the maximum volume of CET is actually determined by policies, and the influence of carbon density of the energy sector as well as the GDP are taken into account.

Fig. 2 and **Table 1** show the logical relationship between the various elements in the CET system. The policy sets the carbon emissions limits for each thermal power generator. The expected sales and purchases of power suppliers depend on CET price and CET holdings. When thermal power generators burn fossil fuels, they will have a demand for carbon quotas. Each plant allocates a certain amount of carbon quotas to allow emissions in accordance with the policy. When the demand for carbon quotas increases, the carbon quotas held by thermal power generators will decrease, the expected sales will decrease, and the CET price will increase. The increase in the carbon emissions control cost of thermal power generators reduces profit, resulting in a decrease in investment in traditional energy installation and a slowdown in the growth of traditional energy installation capacity. In contrast, if the carbon quota demand decreases, the reduction in the carbon emissions cost will eventually increase the installation of traditional energy. In this cycle, the demand for carbon quotas of thermal power generators will change to enter a new round of trading cycles and eventually reach a steady state.

3.2.3. TGC market

To achieve the “30-60” carbon-neutral target, China introduced renewable portfolio standards (RPS) and established a TGC market to exchange the environmental benefits of renewable energy in the form of green certificates.

Fig. 3 and **Table 2** show the logical relationship among variables in the TGC system. In the TGC market, renewable energy generators are suppliers of green certificates, and their expected sales are determined by both the current TGC price and supplier holdings. When the TGC price has an upwards trend or the number of green certificates held by renewable energy generators increases, sales are expected to increase, while excess demand will be reduced. The excess demand causes the supply relations to change, resulting in TGC price drops. The decline in prices leads to a decrease in profits and a decrease in the willingness of renewable energy generators to invest in renewable energy, leading to a slowdown in new installation capacity. The TGC market reaches a steady state in the negative feedback cycle. Investors determine the subsequent investment preference according to the existing production capacity and TGC price, which affects the future installed trend and power generation, thus further affecting the new round of expected supply and demand level.

3.2.4. Electricity market

3.2.4.1. Benchmark generation tariff policy. Under government regulation, electricity is connected to the grid at a fixed benchmark price, independent of market supply and demand. The profits of power generators are directly affected by the benchmark generation tariff, and they cannot adjust their production strategies to obtain more profits according to prices.

3.2.4.2. Market-oriented generation tariff policy. With the market reform of the power industry, it can be predicted that the future generation tariff will depend on the supply and demand relationship under the supervision of the government. The total power generation of various power suppliers is electricity supply, which needs to meet the demand for electricity at a certain rate of annual growth (determined by annual GDP growth). In this process, power suppliers can adjust their

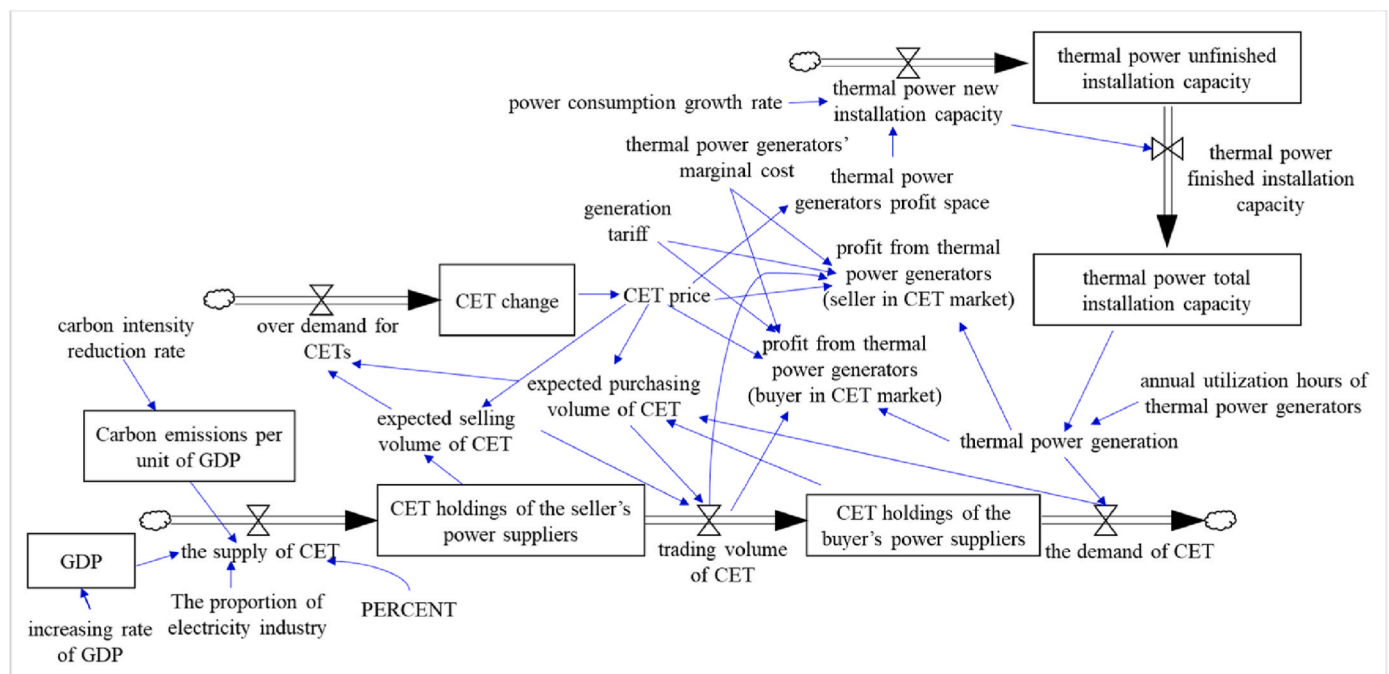


Fig. 2. System dynamics model of the CET market.

Table 1
Variables and formulas in the CET market.

Type	Meaning	Unit	Value/Equation
Constants	increasing rate of GDP	%	0.0573 [29]
	the proportion of electricity industry	%	0.4 [30]
	carbon intensity	%	-0.0064 [31]
	reduction rate		
	thermal power generators' marginal cost	yuan/MWh	0.27 [32]
	annual utilization hours of thermal power generators	h/month	4340/12 [33]
	GDP	100 million	1.01599e+06/12
			+1.01599e+06*increasing rate of GDP/12
	CET holdings of the seller's power suppliers	billion ton	the supply of CET-trading volume of CET
	CET holdings of the buyer's power suppliers	billion ton	trading volume of CET-the demand of CET
Levels	carbon emissions per unit of GDP	Ton/ten thousand	1.8113+ carbon intensity
	CET change	Yuan/ton	50+over-demand for CETs
	profit from thermal power generators (buyers in CET market)	Ten thousand yuan	thermal power generation*generation tariff-CET price*sales of CET-thermal power generators' marginal cost*thermal power generation
	profit from thermal power generators (sellers in CET market)	Ten thousand yuan	thermal power generation*generation tariff + CET price*sales of CET-thermal power generators' marginal cost*thermal power generation
	thermal power unfinished installation capacity	100 million KW	thermal power new installation capacity-thermal power finished installation capacity
	thermal power total installation capacity	100 million KW	thermal power finished installation capacity
	the supply of CET	billion ton	GDP*the proportion of electricity industry*carbon emissions per unit of GDP/10000*0.166
	over-demand for CETs.	Yuan/ton	IF THEN ELSE((expected purchasing volume of CET-expected selling volume of CET)/expected selling volume of CET>30, 30, IF THEN ELSE ((expected purchasing volume of CET-expected selling volume of CET)/expected selling volume of CET < -30, -30, (expected purchasing volume of CET-expected selling volume of CET)/expected selling volume of CET))
	trading volume of CET	billion ton	IF THEN ELSE(MIN(expected purchasing volume of CET, expected selling volume of CET)< 0, 0, MIN(expected purchasing volume of CET, expected selling volume of CET))
	thermal power new installation capacity	100 million KW	75,110*thermal power generators profit space *power consumption growth rate/12*0.5/4340
Flows	the demand of CET	billion ton	power generation of thermal power*0.78/1000
	thermal power finished installation capacity	100 million KW	DELAY FIXED (new investment capacity of thermal units, 12, 0)

Table 1 (continued)

Type	Meaning	Unit	Value/Equation
Auxiliaries	expected purchasing volume of CET	billion ton	IF THEN ELSE(CET holdings of the buyer's power suppliers>the demand of CET, 0, 50/CET price* (the demand of CET-CET holdings of the buyer's power suppliers))
	expected selling volume of CET	billion ton	CET price*CET holdings of the seller's power suppliers/50
	thermal power generators profit space	-	50/CET price*0.38/generation tariff
	thermal power generation	100 million kWh	thermal power total installation capacity*thermal power generators' annual utilization hours
	CET price	Yuan/ton	IF THEN ELSE (SMOOTH 3I(CET Price change, 3, 50)>300, 300, IF THEN ELSE(SMOOTH3I(CET change,3, 50)<10, 10, SMOOTH3I(CET change,3, 50)))

production strategies according to the real-time prices on the market to obtain more profits. In the SD model, the real-time demand variable is added to simulate the fluctuation of real-time electricity prices caused by supply and demand, and the transaction price is finally determined.

3.2.4.3. The relationship among the electricity market and TGC market, CET market. Fig. 4 and Table 3 show the interaction between the electricity market and the other two markets. When the electricity price and TGC price increase, ① The income of renewable energy generators will rise, so will the profit. And they will increase the investment and installation of renewable energy. Additionally, green power affects the TGC price through the TGC market mechanism, forming a negative feedback loop and finally reaching a steady state. ② The increase in green power will also increase the power supply, and the increase in electricity price will also make the power demand smaller. Therefore, the supply will be greater than the demand, the excess demand will be reduced, and the power price will be reduced to a steady state.

For traditional energy power generation, with the increase in electricity price, the profit obtained by thermal power generators will also increase, resulting in an increase in investment and installation capacity of traditional energy, and an increase in power supply, a decline in electricity price when supply exceeds demand, finally reaching a steady state. In addition, with the increase in CET price, the profits available to thermal power generators are reduced, so the investment and installation capacity of traditional energy will be reduced, and then the CET price will be affected through the CET market mechanism, so that the CET price will fall back to a steady state. Finally, the stable operation of the whole system can be guaranteed.

3.3. Model limitation and applicability

As mentioned above, the impact on the entire electricity market of the parallel implementation of TGCs and CET can be analysed by the proposed interactive system dynamics model. However, it is important to note that the actual market process is highly complex, and the proposed model relies on numerous assumptions, making accurate predictions challenging. The designed interactive system dynamics model is primarily used to simulate the potential market evolution process under various low-carbon policies and conduct correlation analysis of market prices.

Additionally, when analysing the interaction between TGC, CET, and the electricity market, we assume that other external conditions remain unchanged. However, unlike the electricity market that primarily focuses on the power sector, the carbon market encompasses various industries, and thus, the carbon trading situation in other sectors may impact it. On the other hand, in practice, market regulation and the

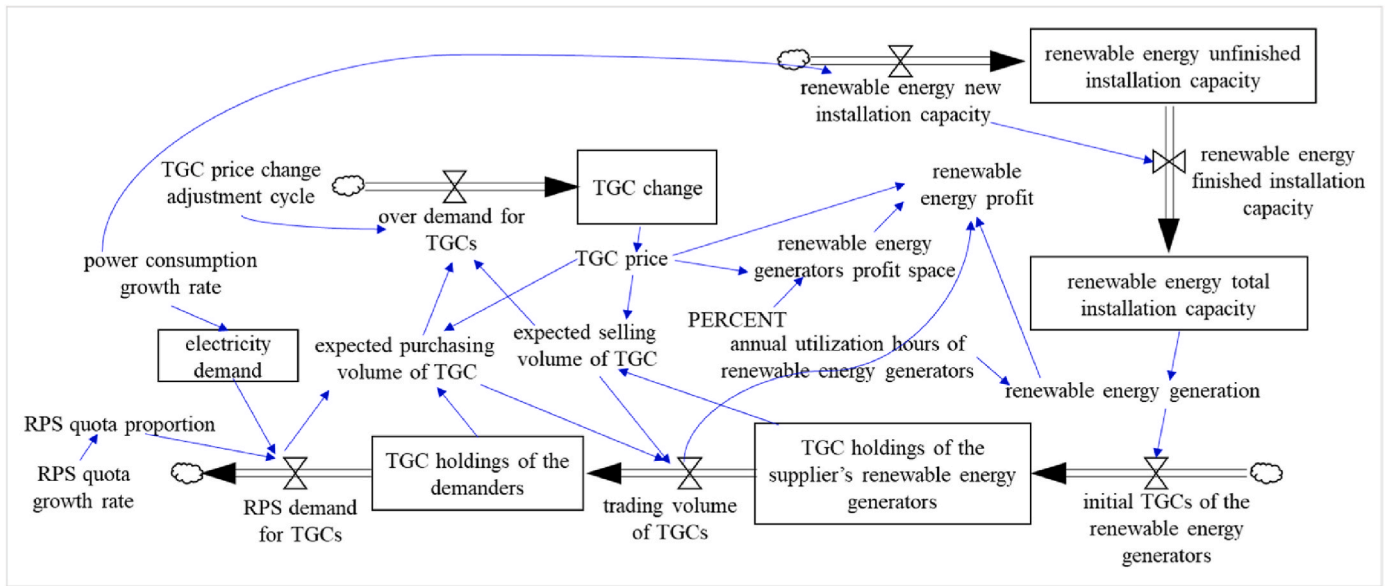


Fig. 3. System dynamics model of the TGC market.

investment risk preferences of various market participants significantly influence the potential effectiveness of low-carbon policies, aspects that are not extensively addressed in the proposed model.

Despite relying on certain assumptions and simplifying certain market processes, the proposed interactive system dynamics model captures two crucial aspects in evaluating low-carbon transformation policies: the price linkage mechanism and the transfer of carbon emissions costs across multiple markets. By adjusting boundary conditions and configuring cost parameters, our model is capable of simulating the interaction between TGC, CET, and the electricity market across various regions, facilitating effective collaboration among different low-carbon policies within diverse market environments.

4. Results and discussion

4.1. Market analysis

4.1.1. TGC market simulation

According to the SD model established above, we set the parameters based on China's actual situation in 2020. The variation trend of the green certificate prices under the consideration of TGC only and both the TGC and CET implementation are illustrated by Fig. 5(a) and (b), respectively. Meanwhile, the relevant data of TGC trading volume, i.e., trading volume and expected sales are illustrated by Fig. 6 (a) and Fig. 6 (b), respectively.

We can observe the price trend from the figure, in early 2020, due to the construction time of the new units, the first batch of units had not been put into operation, which could not affect the TGC market. Simultaneously, the actual trading volume in the TGC market was at a low level, and the trading expectations of both buyers and sellers were not strong, so the TGC price was stable around the initial value. From 2022 to 2023, the TGC price will increase steadily due to the implementation of RPS, which means that power generators need to purchase enough green certificates to meet the policy requirements. Due to the construction time, the electricity construction of renewable energy cannot match the demand level in time, and the expected purchase will be significantly greater than the expected sales. renewable energy generators will tend to hold part of the green certificates and expect the TGC price to rise. Demand in the market will be greater than supply for a period of time, so the TGC price will continue to rise until it reaches the upper limit stipulated by the policy. During this period, the green certificates trading volume will continue to rise, and the TGC market will

grow steadily.

While the TGC price may fall rapidly from 2026 to 2027, the income of renewable energy generators comes from the sum of electricity price income and green certificate income. The high TGC price in the previous stage will make renewable energy generators profitable and have greater investment preferences to expand the installation of renewable energy. At this stage, the installation of some early investment will be completed, and the supply level will be improved. Demand for green certificates from power generators will fall because quotas are met, resulting in prices falling. Expected sales will increase rapidly in the period, renewable energy generators will no longer be willing to hold green certificates in hand, and the actual trading volume will increase significantly. The above downward trend will not last long. From 2027 to 2030, the TGC price trend will become stable and show a growth trend again, and finally remain at the price ceiling stipulated by the policy. This is because the expected RPS quota will reach a high proportion by 2030, and the growth rate of the RPS quota will exceed the growth rate of power demand. The green certificates produced by new units cannot meet the high proportion of RPS quota demand in the later period. Demand will be greater than supply, and the TGC price will increase.

4.1.2. CET market simulation

In the system dynamics model established above, we can obtain the change trend of the CET price under the consideration of CET only and both the TGC and CET implementation, as illustrated by Fig. 7(a) and (b), respectively. There are similarities in the overall trend of price changes between the two, and there are differences in some local trends.

First, the CET price rose in the short term from an initial benchmark price of 50 yuan/ton to approximately 200 yuan/ton and then fell back to approximately 150 yuan/ton. After a period of small fluctuations, equilibrium was eventually reached. It can be found that the curve under the TGC and CET coupling mechanism has more volatility than that under the CET mechanism only, and the price under the equilibrium state is higher because the dual mechanism can better reflect the real-time supply and demand level and environmental value of carbon emissions.

The reasons of the CET price fluctuation can be obtained from Fig. 7 (a). In the first six months of 2021, the initial CET holdings of the thermal power generators were sufficient, resulting in no need to buy CET, and they held a negative attitude to participate in CET market, and the CET price was maintained near the benchmark price of 50 yuan/ton.

Table 2
Variables and formulas in the TGC market.

Type	Meaning	Unit	Value/Equation
Constants	RPS quota growth rate	%	0.0011 [34]
	TGC price change adjustment cycle	Month	3
	annual utilization hours of renewable energy generators	h/month	2147/12 [35]
Levels	RPS quota proportion	–	RPS quota growth rate
	TGCs holdings of the demanders	100 million	trading volume of TGCs-RPS demand for TGCs
	TGCs holdings of the supplier's renewable energy generators	100 million	IF THEN ELSE((initial TGCs of the renewable energy generators-trading volume of TGCs) < -20, -20, initial TGCs of the renewable energy generators-trading volume of TGCs)
	TGC Change	Yuan	over-demand for TGCs
	renewable energy unfinished installation capacity	100 million KW	renewable energy new installation capacity-renewable energy finished installation capacity
	renewable energy total installation capacity	100 million KW	renewable energy finished installation capacity
	renewable energy profit	Ten thousand yuan	renewable energy generation*generation tariff + TGC price*trading volume of TGCs-renewable energy generation*0.504
	RPS demand for TGCs	100 million	RPS quota proportion*electricity demand
	trading volume of TGCs	100 million	IF THEN ELSE(MIN(expected selling volume of TGC, expected purchasing volume of TGC)<0, 0, MIN(expected purchasing volume of TGC, expected selling volume of TGC))
	initial TGCs of the renewable energy generators	100 million	renewable energy generation
Flows	over-demand for TGCs	–	IF THEN ELSE((expected purchasing volume of TGC-expected selling volume of TGC)/100000/settling time of TGC price variation < -0.2, -0.2, IF THEN ELSE((expected purchasing volume of TGC-expected selling volume of TGC)/settling time of TGC price variation/100000>0.2, 0.2, (expected purchasing volume of TGC-expected selling volume of TGC)/TGC price change adjustment cycle/100000))
	renewable energy new installation capacity	100 million KW	75,110*power consumption growth rate/12*0.5*renewable energy generator profitability/2147
	renewable energy finished installation capacity	100 million KW	DELAY FIXED(renewable energy new installation capacity, 12, 0)
	expected purchasing volume of TGC	100 million	IF THEN ELSE(TGCs holdings of the demanders > RPS demand for TGCs, 0, 0.4214*(RPS demand for TGCs-TGCs holdings of the demanders)/TGC price)
	expected selling volume of TGC	100 million	TGC price*TGCs holdings of the supplier's renewable energy generators/0.4214
Auxiliaries			

Table 2 (continued)

Type	Meaning	Unit	Value/Equation
	TGC price	Yuan	IF THEN ELSE(SMOOTH3(TGC Change, 2, 0.4214)>2, 2, IF THEN ELSE(SMOOTH3(TGC Change, 2, 0.4214)<0.1, 0.1, SMOOTH3(TGC Change, 2, 0.4214)))
	renewable energy generators profit space	–	TGC price/0.4214*generation tariff/0.38
	renewable energy generation	100 million kWh	renewable energy total installation capacity*renewable energy generators' annual utilization hours

From the second half of 2021 to the first half of 2023, the price curve will be in a steady upwards stage. The power generation costs and electricity prices of thermal power generators will rise due to the implementation of CET. When the cost of purchasing carbon quotas is lower than the cost of adopting carbon emissions reduction measures, thermal power generators will tend to buy carbon quotas in CET market. With the increase in power demand, thermal power generators will continue to invest in traditional energy generation, and production capacity will continue to improve. As a result, the demand for carbon quotas will be gradually strengthened, eventually leading to an imbalance between supply and demand. Then, the demand will exceed the supply, and the CET price will rise to approximately 200 yuan/ton.

From 2023 to 2025, in the high price stage. With the increase in power demand, the supply of carbon quotas will increase compared with the previous period. The level of supply and demand will reach a relatively stable state with small fluctuations, which reflect changes in the real-time supply and demand balance, but the overall curve will remain near 200 yuan/ton. For thermal power generators, the cost of purchasing carbon quotas will continue to rise, and the profit space will be compressed. In 2026, the rapid decline stage. In the previous high price stage, some generator manufacturers considered risk-resisting measures and invested renewable energy to replace the cost of purchasing carbon quotas. In the process of the accumulation of installation of renewable energy, there will be a point at which the cost of reducing emissions by choosing renewable energy to meet carbon quotas is lower than purchasing carbon quotas in the CET market. At this point, the supply of carbon quotas will exceed demand in a short time, so the CET price will plummet. From 2027 to 2030, in the stable stage, generator manufacturers' investment in thermal power will be stable, and the final CET price will be maintained at 160 yuan/ton to reach the equilibrium price, accompanied by a small fluctuation.

In Fig. 7(b), the trend of the price curve under the TGC and CET coupling mechanism is similar to that under the CET mechanism only. The CET market directly affects thermal power installation and the prices of the electricity markets, and indirectly affects the TGC price by the interaction of the TGC market. In contrast, the TGC market directly affects renewable energy installation and thus on carbon quotas. Under the coupling mechanism, the integration of renewable energy is relatively higher, and the period of high CET price is shorter. This is because the RPS policy forces new renewable energy installation capacity so that the demand for carbon quotas is reduced compared with the CET mechanism only. The high price platform stage becomes shorter, and the curve has greater volatility because the dual mechanism can better reflect the actual environmental value of carbon emissions.

4.1.3. The impact on the electricity market

In Fig. 8, electricity price is influenced by the change in TGC price and CET price under the TGC and CET coupling mechanism. In 2021, the initial electricity price is 0.76 yuan/kWh. In 2022, the electricity price will decrease to 0.69 yuan/kWh. From 2023 to 2026, it will rise rapidly to 2.61 yuan/kWh. Then, it will remain stable at approximately 2.60

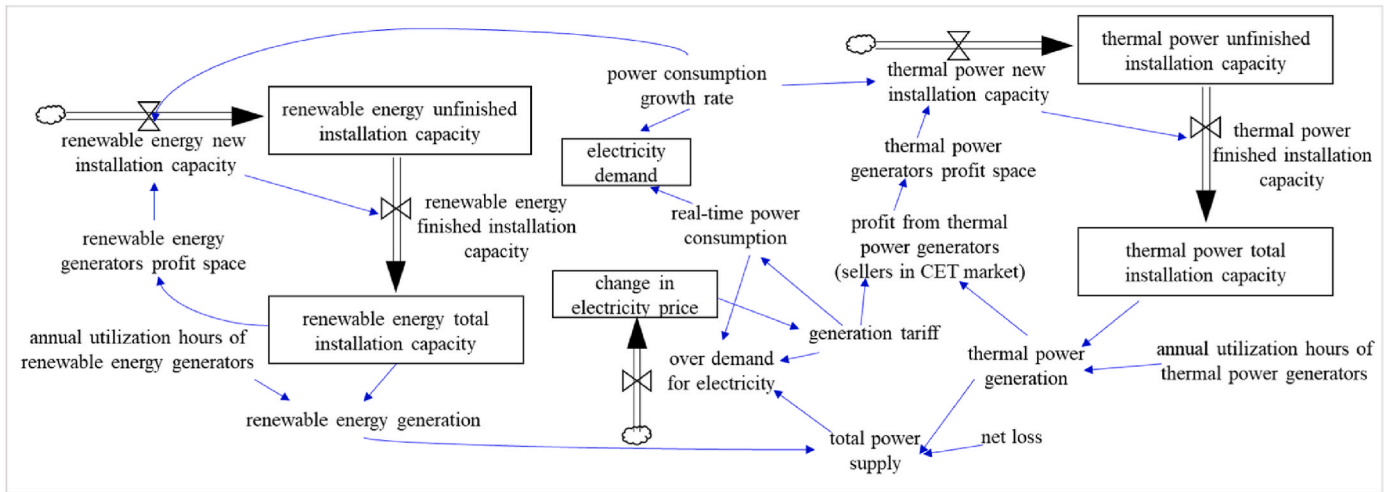


Fig. 4. System dynamics model of the relationship among the electricity market and the other two markets.

yuan/kWh with only minor fluctuations. The results indicate that with the increasing proportion of renewable energy, the TGC mechanism has the most important impact on electricity price, and it is also affected by CET mechanism to produce a small range of fluctuations. The trend indicates that the TGC mechanism can exert higher influence on electricity price than the CET mechanism (especially due to the high proportion of RPS policy).

To further investigate the effect of multi-low carbon policies on the investment behaviour of different types of generation, the installed capacity and total out of green power plants and thermal power plants are given in Fig. 9(a), (b) and Fig. 10(a), (b), respectively. As shown in Fig. 9(a) and (b), for green power, a new installed capacity will appear in the bottleneck and decline stage, and power generation will gradually enter a slow growth stage later under the TGC mechanism only. Compared with the CET and TGC coupling mechanism, the low-carbon target becomes more stringent, which promotes the environmental value of green power. With the development of green power generation, the installed capacity under construction and power generation will continue to increase. Fig. 10(a) and (b) show that for traditional energy, the coupling effect of the dual mechanism brings more stringent emissions reduction pressure to thermal power generators. Compared with the single mechanism, thermal power generators have to undertake higher carbon emissions reduction costs, which leads to slower growth rates.

4.2. Impacts of the TGC and CET coupling mechanism

4.2.1. Impacts on the generation structure

China implements the TGC and CET dual mechanisms to achieve the “30-60” carbon-neutral target. The expected results obtained by simulation are shown in Fig. 11. One can observe that, under the coupling mechanism, the power generation of thermal power generators and renewable energy generators reached 492.262 billion kWh and 939.688 billion kWh, accounting for 34.38% and 65.62% of the total power generation, respectively. The installed capacity of renewable energy is 1.361 billion kw more than the target of 1.2 billion kw, and the thermal power generation and green power generation are increasing year by year, but green power is growing faster than thermal power. Besides, the proportion of thermal power generation is decreasing each year, while the proportion of green power generation is increasing each year. Gradually, green power assumes the main responsibility for power generation so that the energy generation structure is adjusted according to the expected goal. It can be concluded that when the TGC and CET market mechanisms are implemented, the energy structure on the power generation side will be optimized, and the proportion and installed capacity of renewable energy generation will be significantly improved to

achieve the “30-60” carbon-neutral target under the thermal power generators’ emissions reduction requirements and renewable energy generators’ installed capacity requirements.

4.2.2. Impacts on the carbon emissions reduction

To achieve the “30-60” carbon-neutral target, carbon dioxide emissions per unit of GDP in 2030 would be reduced by 60–65% compared with 2005. China’s carbon dioxide emissions per unit of GDP in 2005 were 2.9633 ton/million yuan, which means that it should be controlled at 1.0433 ton/million yuan in 2030 to achieve the desired goal. Based on the established system dynamics model, carbon dioxide emissions are the product of thermal power generation and carbon dioxide emissions per unit of power generation. It is assumed that other industries with carbon emissions also implement CET mechanisms and get the same results with the electricity industry under the parameter settings in this paper. According to the proportion of carbon dioxide emissions in electricity industry, the total national carbon emissions can be speculated.

Under the coupling implementation of CET and TGC the total carbon emissions (the supply of CET) and unit carbon emissions of the electricity industry are shown in Fig. 12. Trends in the figure show that carbon emissions per unit of GDP will fall significantly to 1.043 ton/million yuan in 2030, while carbon dioxide emissions in electricity industry will continue to rise over the decade, but its growth rate will slow significantly to the peak. It can be speculated that the carbon emissions reduction in other industries is expected to achieve the “30-60” carbon-neutral target.

Fig. 13 shows that the total carbon emissions under the CET mechanism are only 5.513 billion tons, while under the coupling mechanism, they are 4.598 billion tons. The total carbon emissions under the coupling mechanism are significantly lower than those under the single CET mechanism, and the curve growth rate is better controlled before the peak. It can be concluded that under the TGC and CET coupling mechanism, the generation structure is effectively optimized to control the carbon dioxide emissions in the energy sector. Assuming that other industries with carbon emissions also implement the CET mechanism and have the same effect, the carbon emissions per unit of GDP will be reduced. Therefore, the coupling market mechanism of TGC and CET contributes to the “30-60” carbon-neutral target.

4.3. Impacts of parameter uncertainty under TGC and CET coupling mechanism

As mentioned earlier, the fundamental purpose of different low-carbon mechanisms, such as Tradable Green Certificates (TGC) and

Table 3
Variables and formulas in the electricity market.

Type	Meaning	Unit	Value/Equation
Constants	power consumption growth rate	%	0.057
	annual utilization	h/month	2147/12 [35]
	hours of renewable energy generators		
	generation tariff	Yuan	0.38 [36]
	annual utilization	h/month	4340/12 [37]
Levels	hours of thermal power generators		
	renewable energy unfinished	100 million	renewable energy new installation capacity-renewable energy finished installation capacity
	installation capacity	KW	
	renewable energy total installation capacity	100 million KW	renewable energy finished installation capacity
	electricity demand	100 million kWh	$75,110 \times \text{electricity demand growth rate}/12$
	change in electricity price	yuan	over-demand for electricity
	thermal power unfinished	100 million	thermal power new installation capacity-thermal power finished installation capacity
	installation capacity	KW	
	thermal power total installation capacity	100 million KW	thermal power finished installation capacity
Flows	renewable energy new installation capacity	100 million KW	$75,110 \times \text{power consumption growth rate}/12 \times 0.5 \times \text{renewable energy profitability}/2147$
	renewable energy finished installation capacity	100 million KW	DELAY FIXED(renewable energy new installation capacity, 12, 0)
	thermal power new installation capacity	100 million KW	$75,110 \times \text{thermal power generators profit space} \times \text{power consumption growth rate}/12 \times 0.5/4340$
	thermal power finished installation capacity	100 million KW	DELAY FIXED(thermal power new installation capacity, 12, 0)
	over-demand for electricity	–	generation tariff*("Real-time power consumption"-the supply of electricity)/"Real-time power consumption"/50
			TGC price/0.4214*generation tariff/0.38
	renewable energy generators profit space	–	
	renewable energy generation	100 million kWh	renewable energy total installation capacity*renewable energy generators' annual utilization hours
	real-time power consumption	100 million kWh	electricity demand*(sale price/0.38)*(-0.25)
	thermal power generators profit space	–	50/CET price*0.38/generation tariff
Auxiliaries	total power supply	100 million kWh	(renewable energy generation + thermal power generation)* (1-net loss)
	thermal power generation	100 million kWh	thermal power total installation capacity*thermal power generators' annual utilization hours

Carbon Emission Trading (CET), is to facilitate the transformation of renewable energy in the power industry, achieving this objective with low cost and high efficiency. However, in practice, technological advancements and regional boundary differences introduce parameter uncertainty, potentially influencing the effectiveness of low-carbon policies. This section delves into the impact of parameter uncertainties on multiple market prices and coupling effects, considering three critical aspects: 1) supply and demand uncertainty, 2) cost characteristics, and

3) generation construction cycles.

4.3.1. The impact of supply and demand uncertainty on market prices

The electricity market and carbon market share a common goal of achieving effective low-carbon development. The impact of carbon emission rights and green power trading on market prices is predominantly manifested in the supply and demand dynamics within their respective markets. In the carbon market, the internalization of carbon dioxide emissions costs through the carbon emission rights market narrows the gap between the costs of electricity sourced from new energy and fossil fuels. Policy makers can encourage power enterprises to engage in carbon market trading by reducing their free carbon quotas, thus stimulating enthusiasm among enterprises to lower emissions and preventing a significant rise in social costs due to emission reduction requirements. To investigate the impact of uncertainties in the ratio of free carbon emission quotas on market prices, Fig. 14 illustrates the evolution of CET prices under different free carbon emission quota ratios.

The CET prices exhibit clear cyclical characteristics. However, a noticeable regularity emerges in the initial and final stages, denoted by blue and pink backgrounds in the figure, as the proportion of free carbon quotas changes. In the initial stage, a decrease in the proportion of free quotas results in a significant acceleration of carbon market price growth. Similarly, in the final stage, particularly after the 8th year, higher carbon quota allocation ratios lead to relatively lower carbon emission prices, consistent with the initial stage. This occurs because when free quotas are scarce, power companies are compelled to engage in carbon market trading. Consequently, the heightened demand for carbon emission rights augments both market liquidity and CET prices.

As previously mentioned, there exists a substantial interconnection between the CET and TGCs. The declining trend of carbon emission rights prices will inevitably bolster the competitiveness of traditional thermal power enterprises in the electricity market. To further analyze the future power load growth's uncertainty from the perspective of supply and demand relations, we conducted a sensitivity analysis on market prices, considering different ratios of free carbon emission quotas and the electricity demand growth rate. In order to comprehend the price trends in different markets and their interdependence, we selected four key nodes and presented them in a heat map. The heat map illustrates the price changes in the carbon market and the electricity market, as depicted in Fig. 15 and 16, respectively.

The figure illustrates that the primary determinant of carbon market prices remains the free carbon emission quota ratio, showing minimal sensitivity to fluctuations in electricity demand growth rate. Conversely, the electricity market price is notably influenced by changes in the load growth rate. With an increasing load growth rate, the electricity market price experiences a rise of 10%–50% at various time points, particularly in scenarios with higher free carbon quota allocations. Despite the ultimate lower price of carbon emission rights, it may still lead to an upsurge in the electricity market price. This phenomenon is mainly attributed to the high proportion of Tradable Green Certificates (TGCs) in electricity market prices. Any increase in TGC prices consequently elevates electricity market prices. It is important to note that CET and TGCs are complementary. A reduction in carbon market emission rights scarcity results in the transmission of energy transformation costs to the electricity market. Moreover, since the carbon price is incorporated into power generation costs, fluctuations in carbon price influence the supply and demand dynamics in the electricity market, as well as the corresponding clearing prices. These findings underscore the critical significance of fostering coordinated development between the electricity market and the carbon market.

4.3.2. The impact of cost characteristics of traditional coal-fired units

The coupling effect of carbon electricity market is not only influenced by the supply and demand relationship of each market itself, but also constrained by the cost characteristics of market members

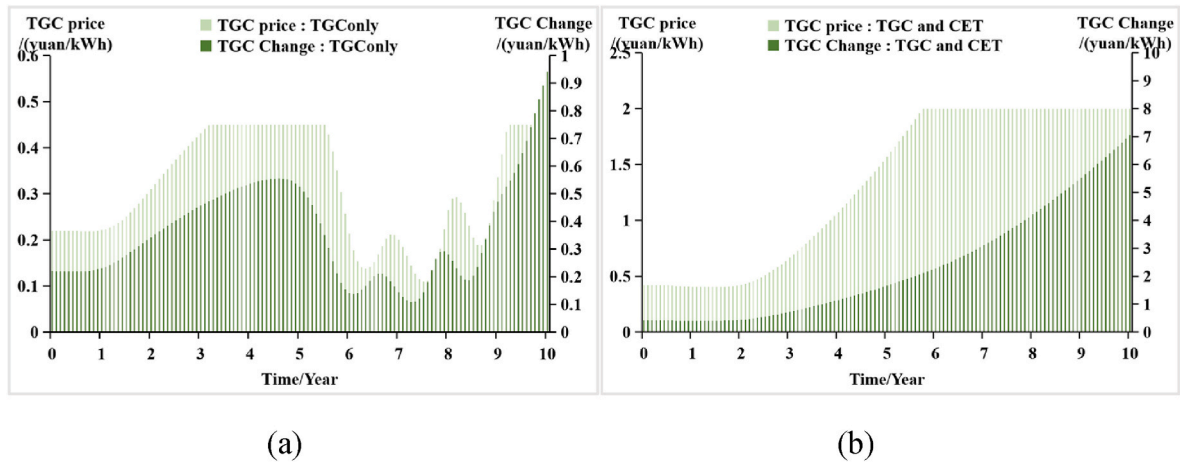


Fig. 5. The variation trend of the TGC prices under different scenarios.

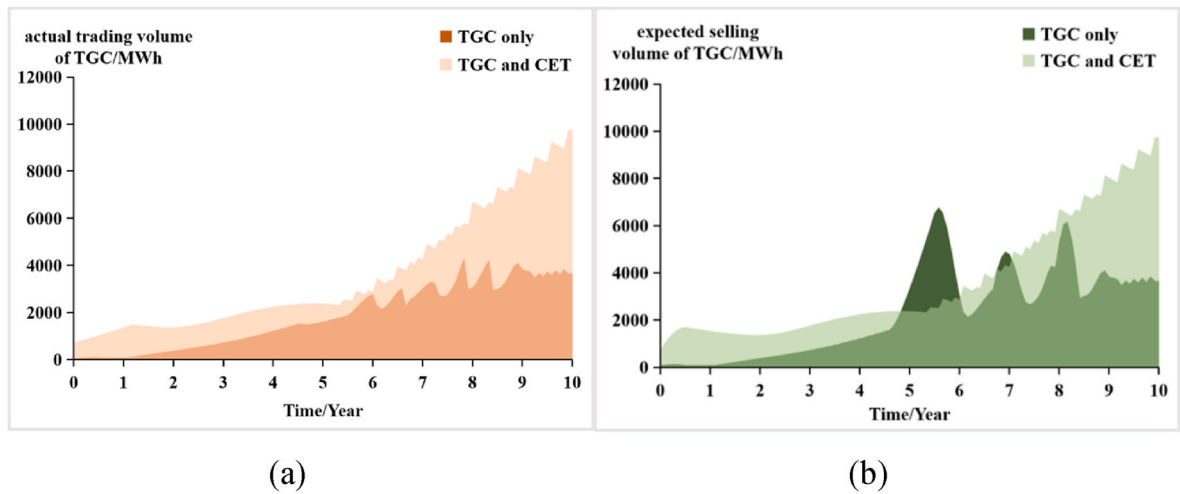


Fig. 6. Actual trading volume and expected selling volume of TGC in different scenarios.

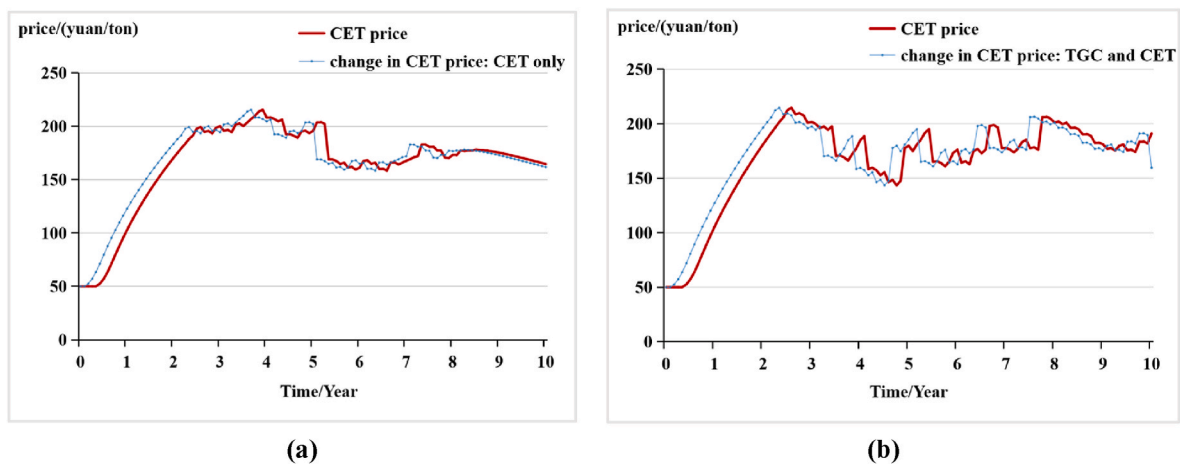


Fig. 7. Carbon emission trading price in different scenarios.

themselves. Especially during the just past COVID-19 epidemic, the energy crisis caused by the rise of coal prices has attracted wide attention. The differences in resource costs will directly affect their market competitiveness, thereby affecting their market share in the energy transformation process. In this subsection, we conduct a sensitivity

analysis on the potential effect of the cost characteristics of traditional coal-fired units. Fig. 17 shows the proportion of newly installed renewable energy capacity and the renewable energy generation under different Levelized Cost of Energy (LCOE) of traditional coal-fired units. Similarly, we presented the initial values and data from four key time

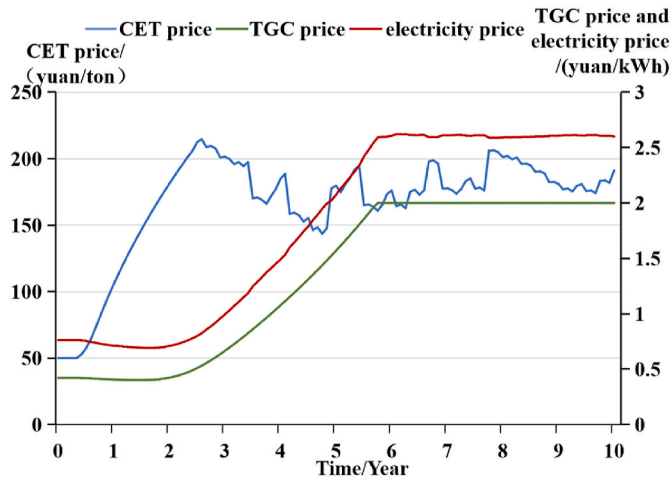


Fig. 8. Three price trends under the coupling mechanism of TGC and CET.

nodes, with different cost scenarios represented by different backgrounds.

The figure demonstrates that, in alignment with the overall energy transformation goal, the uncertainty of thermal power cost parameters does not impact the gradual increase in renewable energy installed capacity and output value. However, a clear observation from the figure reveals that as the LCOE of thermal power gradually rises, the proportion of total installed capacity and power generation from renewable energy experiences a significant upward trend at the same time node. This phenomenon occurs due to variations in the cost of thermal power per kilowatt-hour, which reflects the market competitiveness of

different resource types. An increase in the cost of thermal power per kilowatt-hour weakens its market competitiveness, thereby enabling renewable energy to occupy a higher market share. Further analysis of the trend indicates that an LCOE of 0.3 may serve as the critical point for the competitiveness of these two power generation resources in the market. As the cost of thermal power further increases, the rate of increase in renewable energy market share accelerates.

4.3.3. The impact of different generation construction cycles

Based on the preceding analysis, it is evident that while both renewable energy installation and capacity are exhibiting an increasing trend, the carbon electricity market displays a certain level of periodicity from the perspective of market prices. This periodicity arises due to the delayed regulatory response of various markets and the influence of distinct resource construction cycles.

To examine the impact of construction cycles, a scaling factor “x” is introduced, indicating the ratio of expansion or reduction of the construction cycles for both thermal and renewable energy generation. When $x = 1.0$, it represents the base case, corresponding to the set of renewable energy parameters mentioned above. Additionally, considering the variations in regional power structures and energy transformation processes, we also account for differences in the proportion of initial renewable energy installed capacity. Fig. 18 illustrates the evolution of accumulated installed capacity of renewable energy under different construction cycles, along with the initial renewable energy penetration denoted as η_{RE} . In order to better analyze the changing trends of renewable energy market share in different scenarios, we have chosen a fixed cumulative installed capacity represented by the red dashed line in the figure. The corresponding timeline reflects the time required to achieve this installed capacity target in various scenarios.

From Fig. 18, it can be said that although the trend of installed

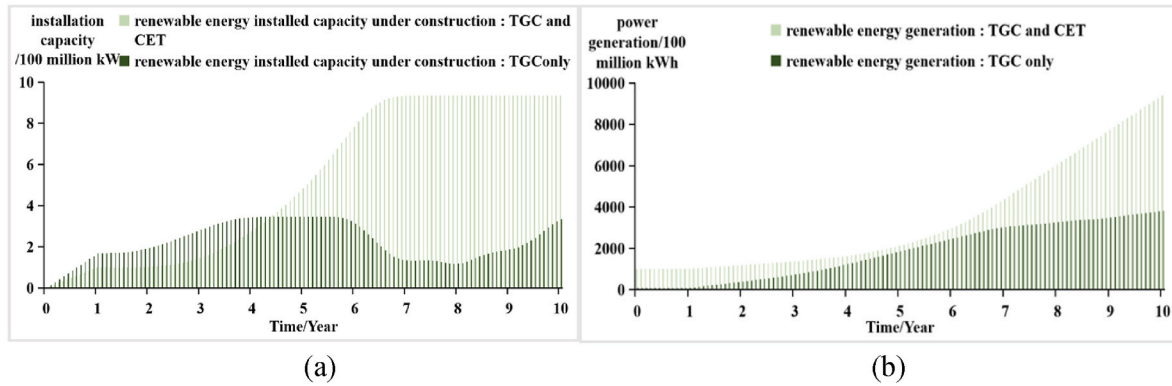


Fig. 9. The installed capacity and total out of green power generators.

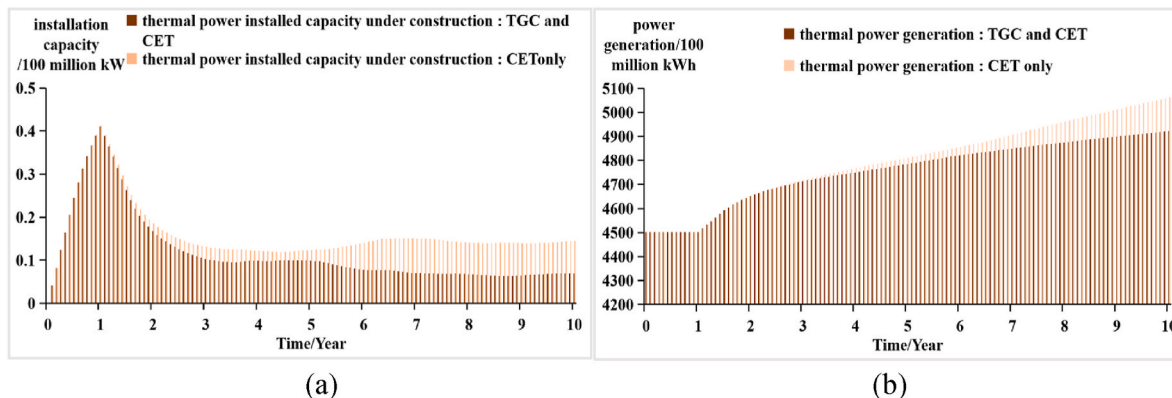


Fig. 10. The installed capacity and total out of thermal power generators.

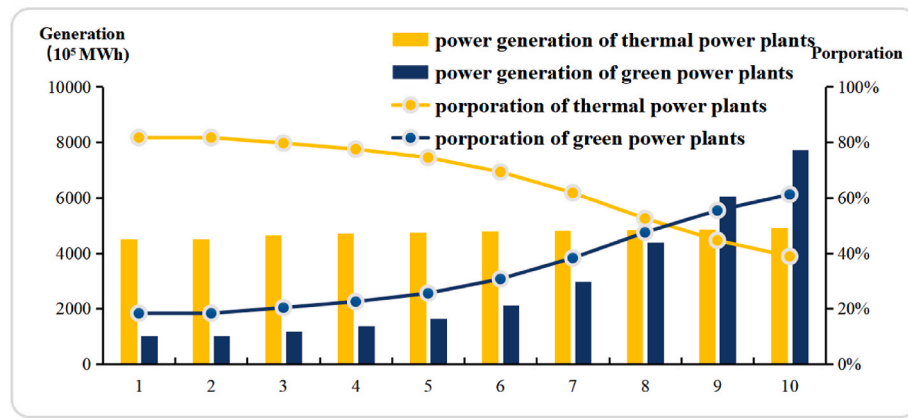


Fig. 11. Power generation of thermal and green power plants under coupling implementation of TGC and CET.

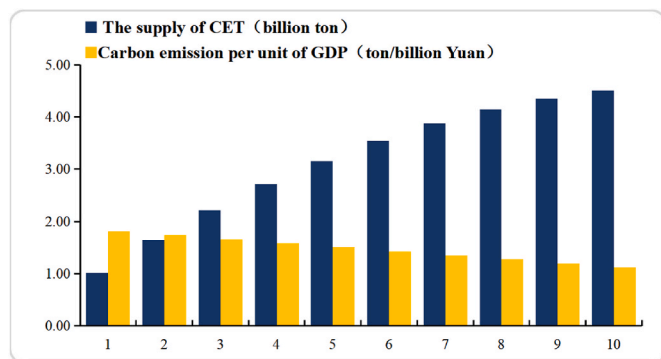


Fig. 12. The supply of CET and carbon dioxide emissions per unit GDP under the coupling mechanism of TGC and CET.

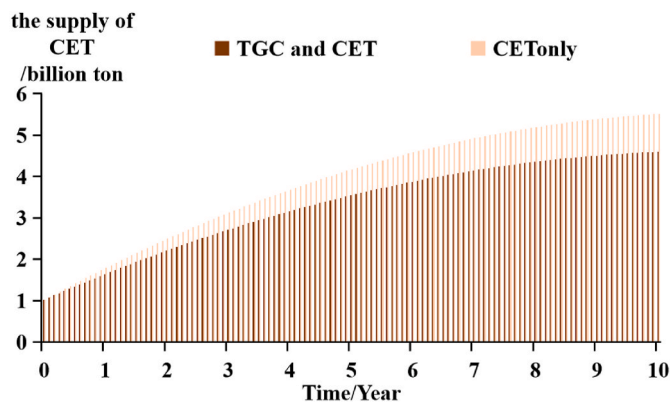


Fig. 13. The supply of CET in different scenarios.

capacity growth of renewable energy is relatively consistent, the shortening of the construction cycle of power generation resources will accelerate the process of energy transformation to varying degrees, namely increasing the market share of renewable energy. In practice, the construction cycle will also include processes such as bidding approval for investment plans, which indicates that policy support and simplification of relevant processes are conducive to energy transformation in the carbon-electricity market. Comparing the results under different initial renewable energy penetration rates, we are surprised to find that a higher initial renewable energy ratio does not necessarily mean a faster energy transformation process. This is mainly because for a certain region, a higher initial renewable energy ratio may lead to sufficient TGC supply, which may reduce the market competitiveness of

renewable energy and instead slow down the subsequent development process. This result is of great significance for policy formulation and low-carbon policy design tailored to local conditions in different regions in practice.

5. Conclusions and policy implications

The objective of this study is to investigate the combined policy effect of TGCs and CET in the carbon-electric coupling market. A system dynamics model is established based on market equilibrium theory to quantitatively analyze the coupling effect of the three markets and compare the impact of these policies on the power supply structure. The actual trading modes of the electricity market under the dual mechanisms of TGC and CET for the next decade are simulated on a monthly time scale. From the perspective of policy implementation effectiveness, the coupling effect among different markets shows a gradual upward trend, and the implementation of TGCs and CET can influence the generation structure. To achieve the dual carbon goals, the total investment capacity of variable renewable energy is projected to reach 1.2 billion kW, and carbon emissions can be effectively controlled to meet the reduction target with unit carbon emissions nearly 70% lower than those in the early 21st century. Under the coupling mechanism, total carbon dioxide emissions are significantly lower than those under the single CET mechanism, with a significantly slower growth rate. The unit carbon emissions are reduced to 1.0433 ton/million yuan. However, it should be noted that the implementation of diverse low-carbon policies may result in higher social costs. The results indicate that the price of the electricity market has substantially increased. Both carbon emission quotas and renewable energy quotas directly impact the supply and demand relationship in their respective markets. For instance, the proportion of renewable energy generation is much higher when considering the TGC market, while the duration of high carbon prices exhibits a decreasing trend.

From the perspective of market behavior, diverse low-carbon policies and the coupling of carbon electricity markets can facilitate power companies in achieving *trans*-formation under emission reduction targets. The example results also demonstrate that changes in the cost of different resources and the construction cycle directly influence the market competitiveness of various power generation resources, as evident from the market share and observed market price signals. Across multiple markets, including CET, TGC, and the electricity market, prices act as crucial signals for market participants in making investment decisions. The transfer of social welfare and conflicts of interest among market players profoundly impacts the operational efficiency of both the electricity market and the carbon market. The evolution results of this study reveal that the equilibrium price formed in various market interactions is influenced not only by the coupling relationship between markets but also by the price ceiling of each market. Additionally, the

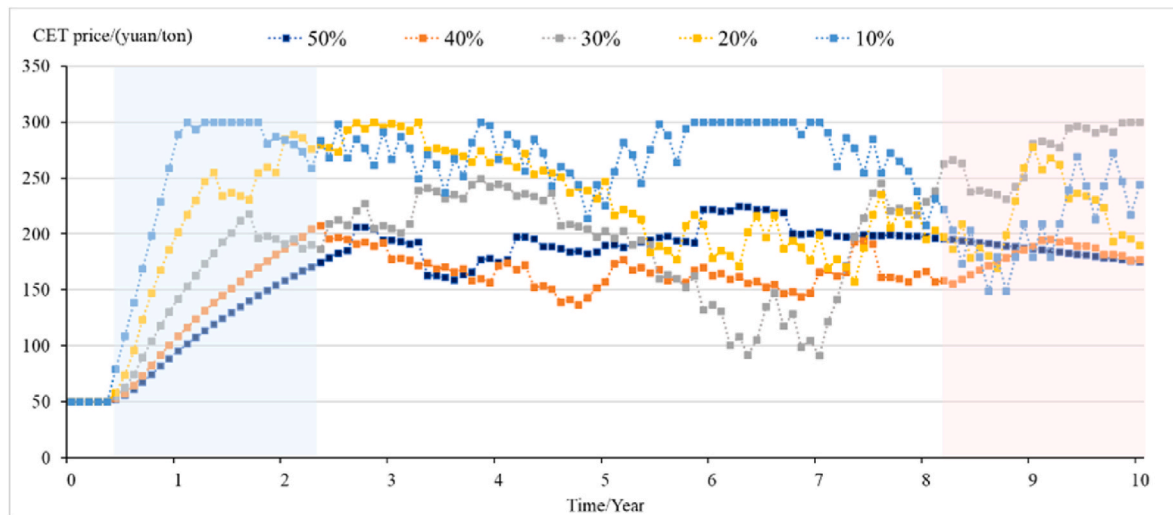


Fig. 14. CET prices evolution under different free carbon emission quota ratio.

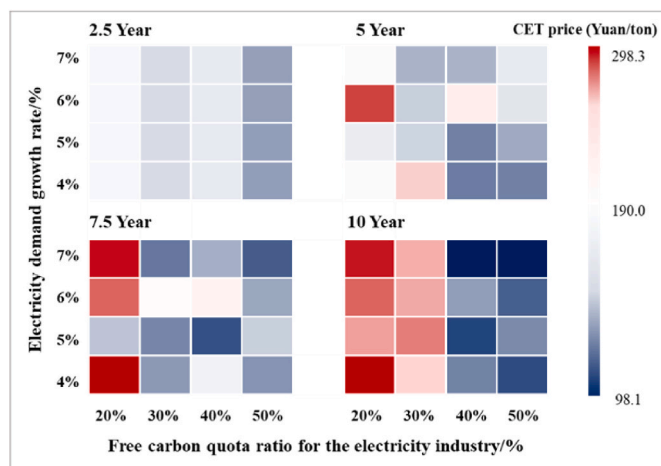


Fig. 15. CET prices free carbon emission quota ratio as well as the electricity demand growth rate.

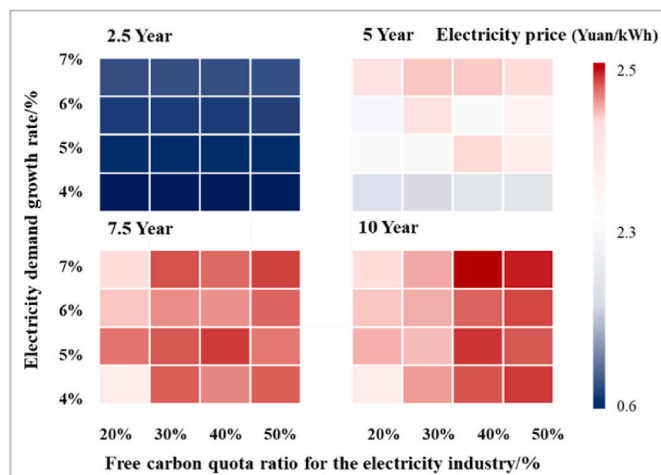


Fig. 16. Electricity prices under free carbon emission quota ratio as well as the electricity demand growth rate.

allocation ratio of free carbon quotas significantly impacts the prices of carbon and electricity markets, as well as the overall allocation of energy transformation costs. Currently, China's carbon market has notably low prices, accounting for only 10% of the EU's carbon emission price. Moreover, the TGC system remains in a pilot state, indicating relatively low levels of marketization. This highlights the necessity of an efficient and orderly regulatory mechanism during the initial stage of China's electricity and carbon market construction. Establishing reasonable market prices for both CET and TGC, along with an adequate free carbon quota allocation ratio, is essential in providing effective and sustainable incentive signals for carbon emission reduction enterprises. To this end, the government should design different renewable energy targets and carbon emission quotas based on energy planning needs, create an efficient and orderly regulatory mechanism at the macro level, and coordinate the development of the carbon market and the electricity market.

In practice, the effectiveness of low-carbon policies is influenced by technological progress and variations in regional boundary conditions, leading to parameter uncertainties. This paper considers the impact of parameter uncertainties on multiple market prices and coupling effects from three perspectives: 1) supply and demand uncertainty, 2) cost characteristics, and 3) generation construction cycles. Additionally, the complementary and coupling effects of Carbon Emission Trading (CET) and Tradable Green Certificates (TGC) vary across different regions, influenced by various parameter uncertainties. While parameter uncertainty does not alter the overall trend of energy transformation, it does affect specific policy outcomes. From the perspective of market mechanism design, the carbon market aims to guide energy-consuming enterprises in conserving energy and mitigating emissions, while the central focus of electricity market reform is to ensure that electricity prices accurately reflect market supply, demand, and the costs of carbon emission reduction. For instance, the CET price declines with an increase in mandatory Renewable Portfolio Standard (RPS) quotas, potentially reducing the effectiveness of the CET system. Conversely, the RPS policy may become redundant if carbon quotas are raised. As mentioned earlier, the electricity market and carbon market exhibit a trend of mutually reinforcing promotion and coordinated development. This once again emphasizes the importance of tailoring low-carbon policies to local conditions, as achieving carbon neutrality requires a comprehensive and profound transformation of economic and social systems. The key to successful energy transformation lies in coordinating variable policies and fostering synergistic interactions among multiple markets. Overall, the government should leverage the complementary advantages of coupling effects in policy combinations and carefully coordinate

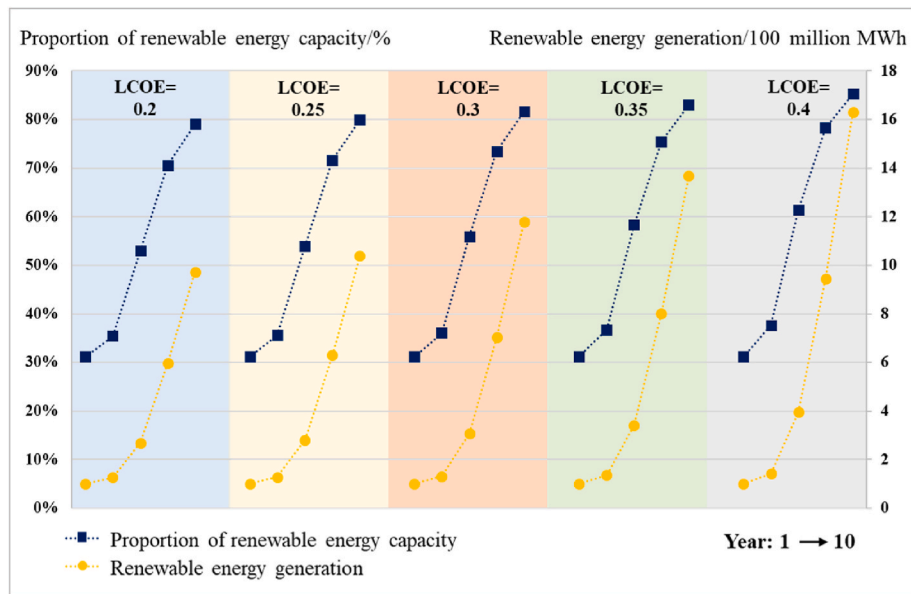


Fig. 17. Proportion of newly installed renewable energy capacity and the renewable energy generation under different LCOE of traditional coal-fired units.

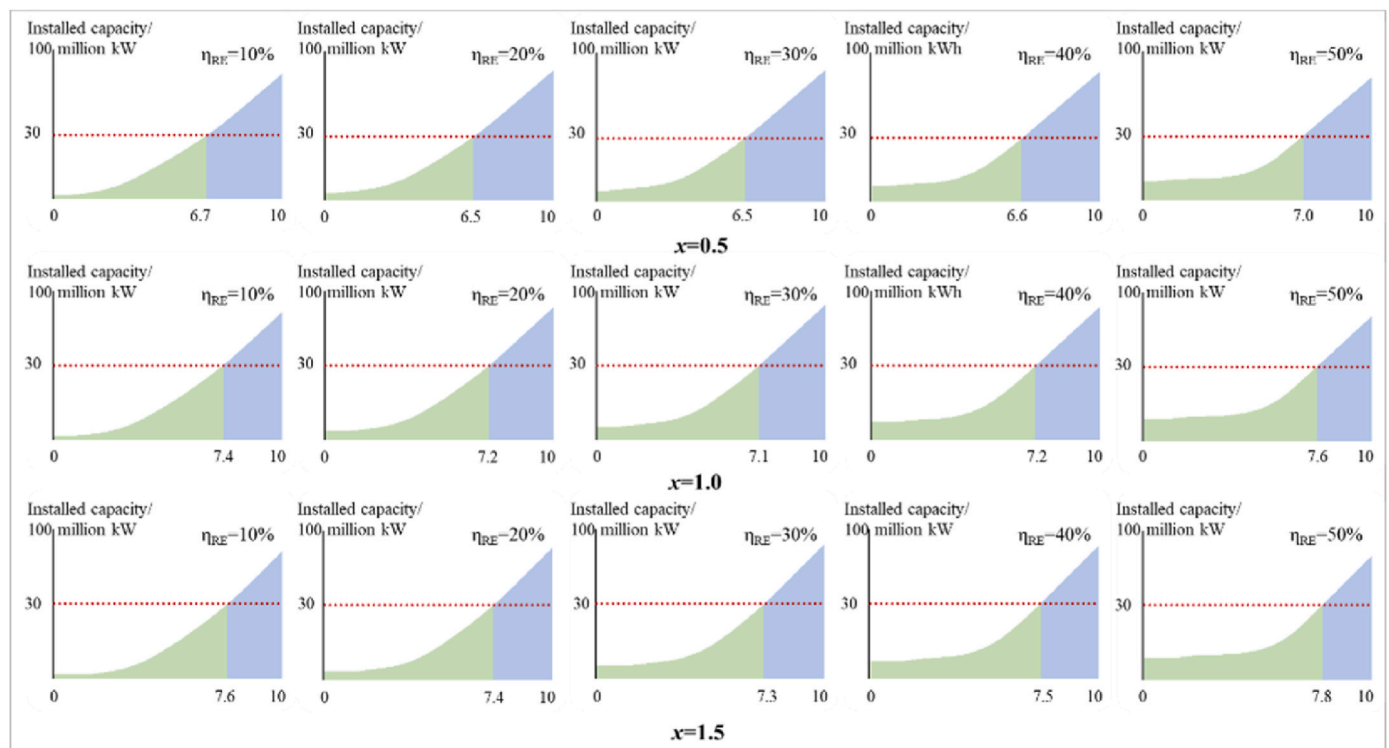


Fig. 18. Accumulated installed capacity of renewable energy under different construction cycles as well as the initial renewable energy penetration.

the application of dual mechanisms based on the specific conditions in different regions.

It should be noted that our proposed interactive system dynamics model made a number of assumptions that future work should address to yield more general results. This would include of the potentially significant impact of investment risk preferences and strategic market behaviours exhibited by different market participants, as well as the influence of market regulation and diverse market designs in different regions. Besides, both the presented results as well as the theoretical analysis have indicated that there may be overlap effects between different low-carbon policies, such as TGC and CET, and unreasonable

policy combinations may lead to higher social costs. The coordination of TGC, CET, and other low-carbon policies to adapt to different regions with varying boundary conditions deserve in-depth study in our future work.

Credit author statement

Xin chang: methodology, investigation, formal analysis, zhaoyuan wu: conceptualization, methodology, jingting wang: software. Xinyu zhang: visualization, data curation. Ming zhou: supervision, writing – review & editing, funding acquisition. Tao yu: project administration,

data curation. Yuyang wang: data curation

Declaration of competing interest

The authors declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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